7.2.4 Software Verification Process

NOTE — The Software Verification Process confirms that software work products and services accurately and completely reflect their specified requirements. Verification may be performed internally or by an independent organization, depending on project criticality, risk, and contractual obligations.

7.2.4.1 Purpose

The purpose of the Software Verification Process is to **confirm that each software work product and/or service of a process or project properly reflects the specified requirements**, ensuring correctness, completeness, and conformance throughout the software life cycle.

7.2.4.2 Outcomes

As a result of successful implementation of the Software Verification Process:

a) a verification strategy is developed and implemented;
b) criteria for verification of all required software work products are identified;
c) required verification activities are performed;
d) defects are identified and recorded; and
e) results of the verification activities are made available to the customer and other involved parties.

7.2.4.3 Activities and Tasks

The project shall implement the following activities and tasks in accordance with applicable organizational policies and procedures with respect to the Software Verification Process.

7.2.4.3.1 Process Implementation

This activity consists of the following tasks:

7.2.4.3.1.1 Verification Strategy Determination

The need and extent of verification shall be determined based on the **criticality** of the software. Criticality may be gauged in terms of:

a) the potential of an undetected error in a system or software requirement to cause death or personal injury, mission failure, or financial or catastrophic equipment loss or damage;
b) the maturity of and risks associated with the software technology to be used; and
c) the availability of funds and resources.

7.2.4.3.1.2 Verification Process Establishment
If the project warrants a verification effort, a verification process shall be established to verify the software product. This process shall cover planning, execution, evaluation, and reporting activities.

7.2.4.3.1.3 Independent Verification
If the project warrants an **independent verification effort**, a qualified organization responsible for conducting verification shall be selected. This organization shall be assured of **organizational independence and sufficient authority** to perform verification activities objectively.

7.2.4.3.1.4 Verification Planning
A **Verification Plan** shall be developed and documented. The plan shall specify:

- activities, methods, techniques, and tools for performing verification tasks (as defined in 7.2.4.3.2);
- the life cycle activities and software products to be verified;
- responsibilities, resources, schedule, and verification levels;
- acceptance and completion criteria for each verification activity;
- reporting mechanisms and escalation paths.

7.2.4.3.1.5 Verification Implementation and Reporting
The verification plan shall be implemented as documented. Problems and non-conformances detected by the verification effort shall be entered into the **Software Problem Resolution Process (7.2.8)**. All identified issues shall be tracked to closure.
Results of the verification activities shall be documented and made available to the

acquirer and other involved organizations.

7.2.4.3.2 Verification

This activity consists of the following tasks:

7.2.4.3.2.1 Requirements Verification
The requirements shall be verified considering the following criteria:

a) system requirements are consistent, feasible, and testable;
b) system requirements have been appropriately allocated to hardware items, software items, and manual operations according to design criteria;
c) software requirements are consistent, feasible, testable, and accurately reflect system requirements;
d) software requirements related to safety, security, and criticality are correct, as demonstrated by suitably rigorous methods.

7.2.4.3.2.2 Design Verification
The design shall be verified considering the following criteria:

a) the design is correct, consistent with, and traceable to requirements;
b) the design implements the proper sequence of events, inputs, outputs, interfaces, logic flow, timing and sizing budgets, and error definition, isolation, and recovery;
c) the selected design can be derived from requirements;
d) the design implements safety, security, and other critical requirements correctly, as shown by suitably rigorous methods.

7.2.4.3.2.3 Code Verification
The code shall be verified considering the following criteria:

a) the code is traceable to design and requirements, testable, correct, and compliant with requirements and coding standards;
b) the code implements the proper event sequence, consistent interfaces, correct data and control flow, completeness, appropriate allocation, timing and sizing budgets, and error definition, isolation, and recovery;

c) the selected code can be derived from design or requirements;
d) the code implements safety, security, and other critical requirements correctly, as shown by suitably rigorous methods.

7.2.4.3.2.4 Integration Verification
The integration shall be verified considering the following criteria:

a) software components and units of each software item have been completely and correctly integrated into the software item;
b) hardware items, software items, and manual operations of the system have been completely and correctly integrated into the system;
c) integration tasks have been performed in accordance with the integration plan.

7.2.4.3.2.5 Documentation Verification
The documentation shall be verified considering the following criteria:

a) documentation is adequate, complete, and consistent;
b) documentation preparation is timely;
c) configuration management of documents follows specified procedures.

7.2.4.4 Deliverables

The following deliverables shall be produced during this process:

- Software Verification Plan;
- Verification procedures, checklists, and tools;
- Requirements, design, code, integration, and documentation verification reports;
- Verification logs and defect reports;
- Independent verification reports (if applicable);
- Traceability matrices linking requirements, design, code, and tests;
- Verification summary report for acquirer or oversight bodies.

7.2.4.5 Audit and Configuration Control

All verification plans, reports, logs, and related artifacts shall be placed under appropriate configuration management.

Audit evidence shall be retained in accordance with organizational and regulatory policies.
Verification evidence shall be digitally signed or otherwise protected to ensure integrity.
Verification baselines shall be updated to reflect resolved defects and approved changes.

Cross-References

This process interacts with the following:

- **Software Requirements Analysis Process (7.1.2)** — provides input requirements to be verified.
- **Software Architectural Design (7.1.3)** and **Detailed Design (7.1.4)** — provide design artifacts for verification.
- **Software Construction (7.1.5)** — provides code and unit artifacts for code verification.
- **Software Integration (7.1.6)** — provides integration results for verification of assembled components.
- **Software Qualification Testing (7.1.7)** — consumes verification results to plan qualification tests.
- **Software Quality Assurance (7.2.3)** — coordinates verification activities for compliance and reporting.
- **Software Validation (7.2.5)** — uses verification evidence to support validation efforts.
- **Software Review (7.2.6)** and **Software Audit (7.2.7)** — review verification outputs.
- **Software Problem Resolution (7.2.8)** — receives defects and non-conformances found during verification for resolution tracking.